ChE 312 – Transport Processes II

Combine equilibrium relations with mass balances.

Semibatch distillation – Balance on the liquid residue.

Continuous stagewise distillation – Coupled balances on stages.

Examine PS03.

3a. For pure benzene and pure toluene, construct a spreadsheet (Excel or Google Sheets) to calculate the vapor pressures and vapor-pressure ratio ($P_{vapor,benzene}$ / $P_{vapor,toluene}$) as temperature varies from the boiling point of pure benzene to the boiling point of pure toluene. Use Antoine's equation for pure-species vapor pressure.

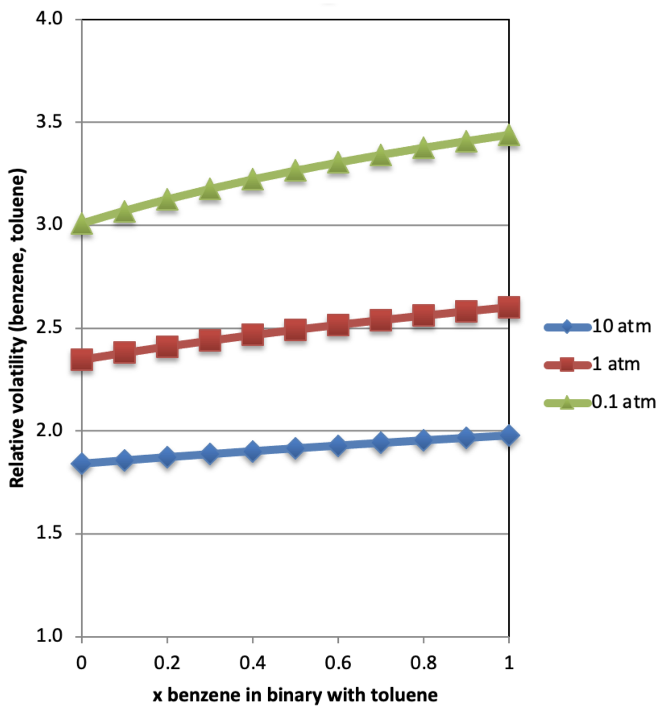
- What is the average $P_{vapor,benzene}$ / $P_{vapor,toluene}$, and how much does the actual ratio vary? That is, how close is benzene-toluene to having "constant relative volatility"?

3b. Constant relative volatility gives a convenient equation for binary equilibrium, but it is an approximation. Construct a spreadsheet to compute a table and plot y vs x (that is, y1 vs x1) for ideal binary vapor-liquid equilibrium with a given constant alpha.

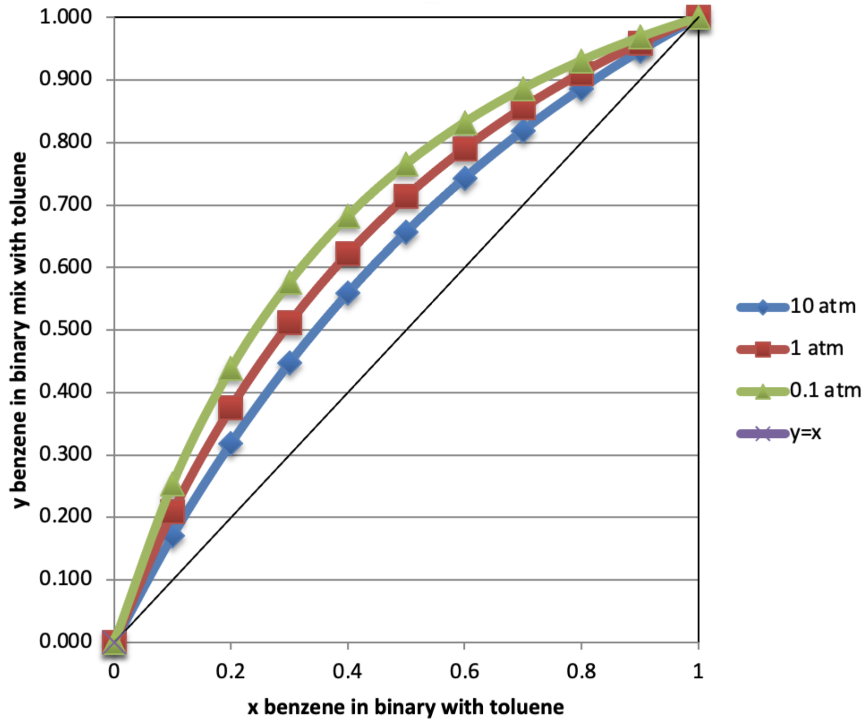
- Note that this calculation is independent of temperature and of pressure.
- Use the α for benzene-toluene binary vapor-liquid equilibrium.
- [You should have a table where x and y both vary from zero to 1.]

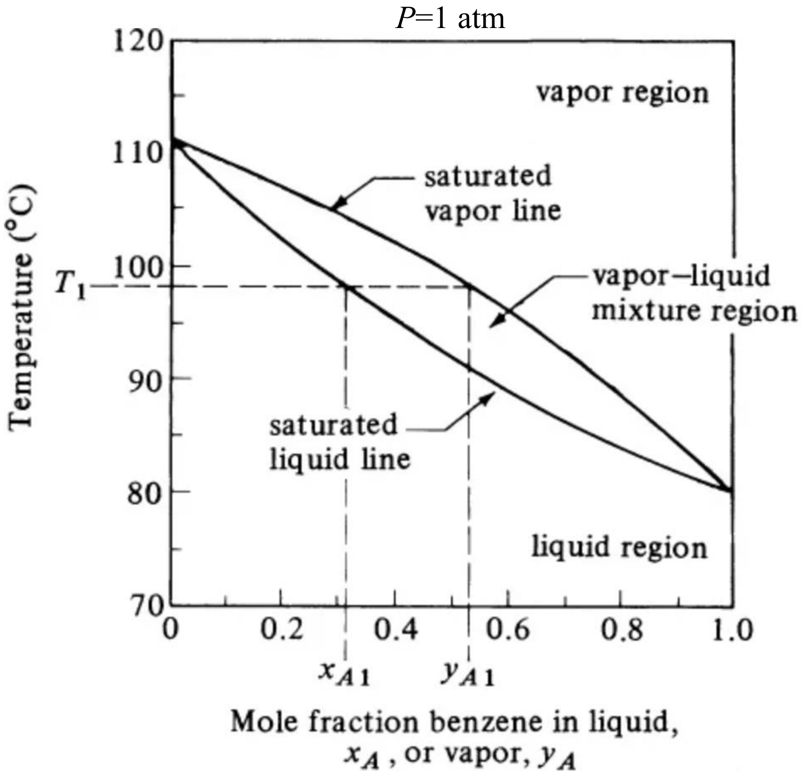
"Constant relative volatility" is useful even if approximate.

- "Relative volatility" $\alpha_{ij}(T) = \frac{y_i/x_i}{y_j/x_j} = \left(\frac{P_{vap,1}}{P_{vap,2}}\right)$;
  $\Rightarrow y(x,T) = \frac{\alpha_{ij}x}{1+(\alpha_{ij}-1)x}$
  - If $\left(\frac{P_{vap,1}(T)}{P_{vap,2}(T)}\right)$ is constant with T, then "constant relative volatility."
- Very convenient when valid, giving $y(x)$ as the T effect is eliminated.
- Usually for molecules with similar interactions, like benzene and toluene.



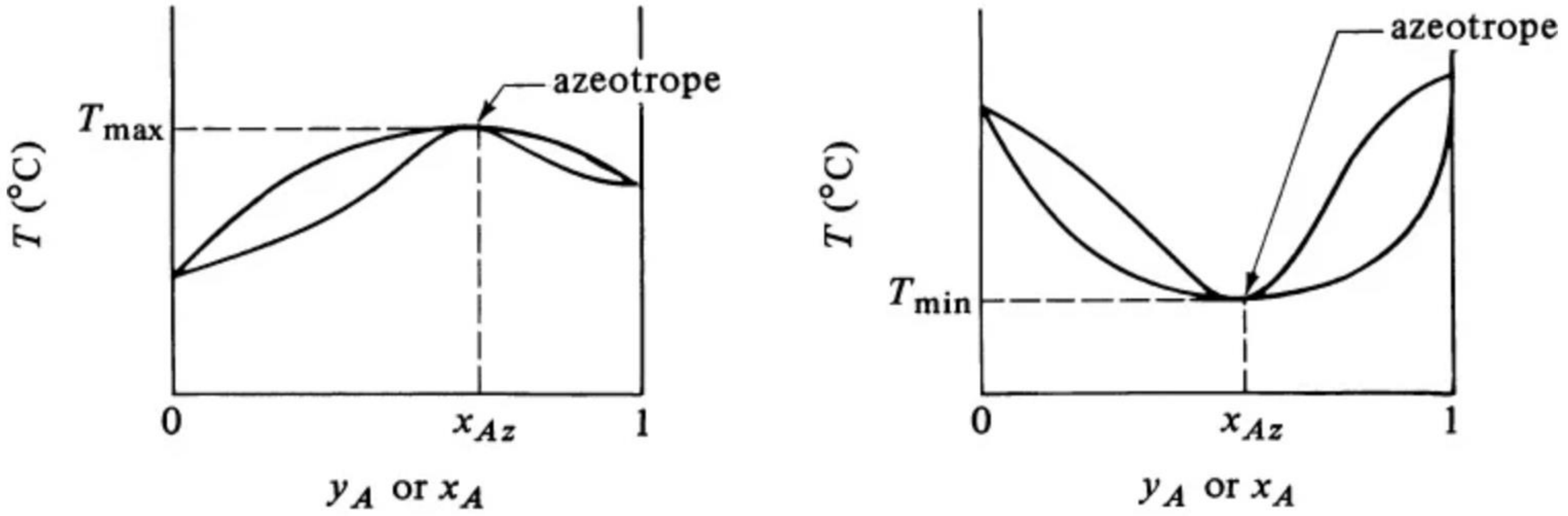
At constant *P*, graph *y* vs. *x* or *T* vs. *x* and *y*.





If the mixture *y* ever equals *x*, we have an azeotrope,

$$y_{Az} = x_{Az}.$$

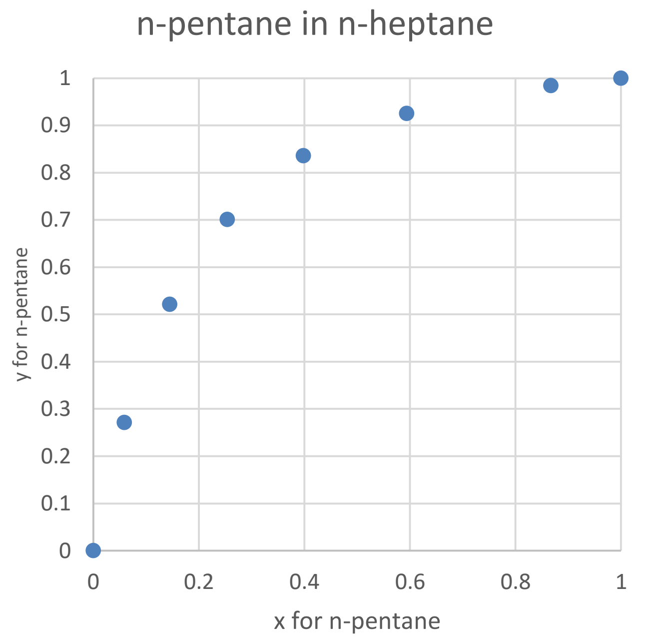


- Then there is no difference between *y* and *x* to use for separation.
  - Unless vapors and liquids are all < x_{Az} or all > x_{Az}.

Example 26.3-2. Simple differential distillation.

- 100 mol initially, 50.0 mol % *n*-pentane mixture with *n*-heptane (should be ideal)
- Differential distillation at P=101.3 kPa until 40 mol are distilled.
- Average composition of total vapor and remaining liquid?

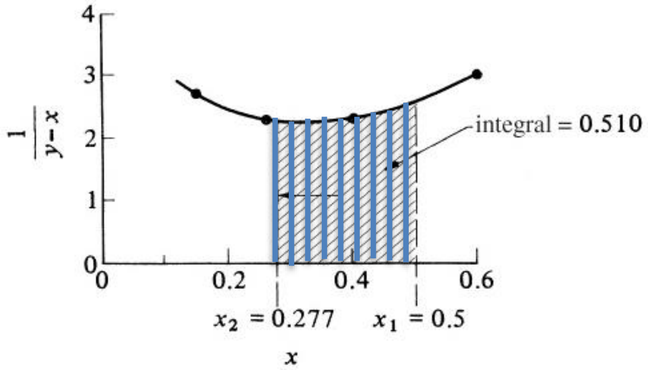
***x_{eq}*, *n*-pentane in *n*-heptane**	***y_{eq}*, *n*-pentane in *n*-heptane**
1.000	1.000
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271
0.000	0.000



Solve using $\frac{dL}{L} = d \ln L = \frac{dx}{y_{eq} - x}$

- Integrate from $L_1 = 100\ mol$ and $x_1 = 0.500$ to $L_2 = 60$ using $ln\frac{L_2}{L_1} = \int_{x_1}^{x_2} \left(\frac{1}{y_{eq}-x}\right) dx$, the Rayleigh equation.
- Then $\ln\frac{L_2}{L_1} = \frac{60}{100} = 0.510.$
- Before-and-after mole balance is: $L_1x_1 = L_2x_2 + (L_1 - L_2)y_{ave}$, giving $y_{ave} = \frac{L_1x_1 - L_2x_2}{L_1 - L_2}$
- Thus, we need to find x_2 from the integration.

- One approach: Integrate using a graph+table of $\frac{1}{y_{eq}-x}$ vs. x as a sum of $\left(\frac{1}{y_{eq}(x)-x}\right)_{i,ave} \cdot (x_{i+1} - x_i)$



Then x_2 =0.277 gives y_{ave} =0.835.

Solve using $\frac{dL}{L} = d\ \ln L = \frac{dx}{y_{eq}-x}$

- Integrate from $L_1 = 100\ mol$ and $x_1 = 0.500$ to $L_2 = 60$ using $ln\frac{L_2}{L_1} = \int_{x_1}^{x_2}\left(\frac{1}{y_{eq}-x}\right)dx$, the Rayleigh equation.
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- Thus, we need to find x_2 from the integration.

- Or fit $y = \frac{\alpha x}{1+x(\alpha-1)}$

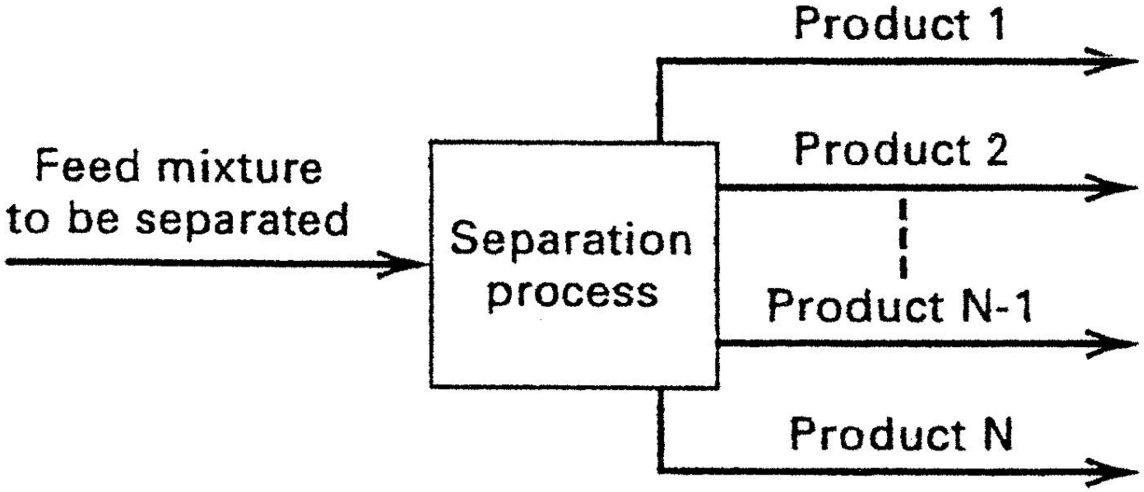
αx/(1+x(α-1))	$(y_{Eq}-y_{Pred})^2$
1.0000	0.0000
0.9774	0.0000
0.9065	0.0003
0.8142	0.0005
0.6929	0.0001
0.5292	0.0001
0.2936	0.0005
0.0000	0.0000
	0.0015
alpha = 6.63	

to get α and compute the integral:

$$\sum\left(\frac{1}{\frac{\alpha x}{1+x(\alpha-1)}-x}\right)_{i,ave} \cdot (x_{i+1} - x_i)$$

using a set of x values.

Now consider multi-tray, continuous distillation.



Study https://www.youtube.com/watch?v=I70jgRpf80o (auf Deutsch)

Fig. 1.5; Seader, Henley, and Roper, *Separation Process Principles,* 3rd Ed.

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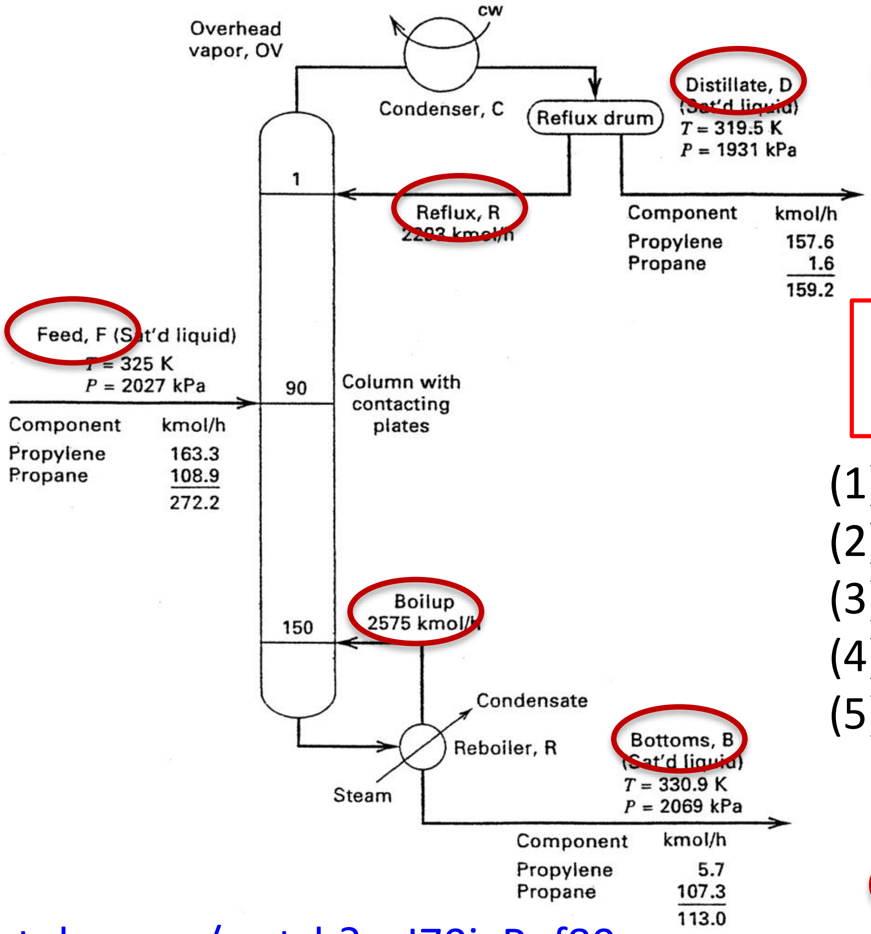
Continuous distillation is a workhorse for separations.

Supply heat thru feed and reboiler; Supply coolant at top to condense and return part of the liquid; Thus countercurrent V-L contacting.



"Reflux ratio"=Reflux/D

What steps to develop a model for analysis or design?

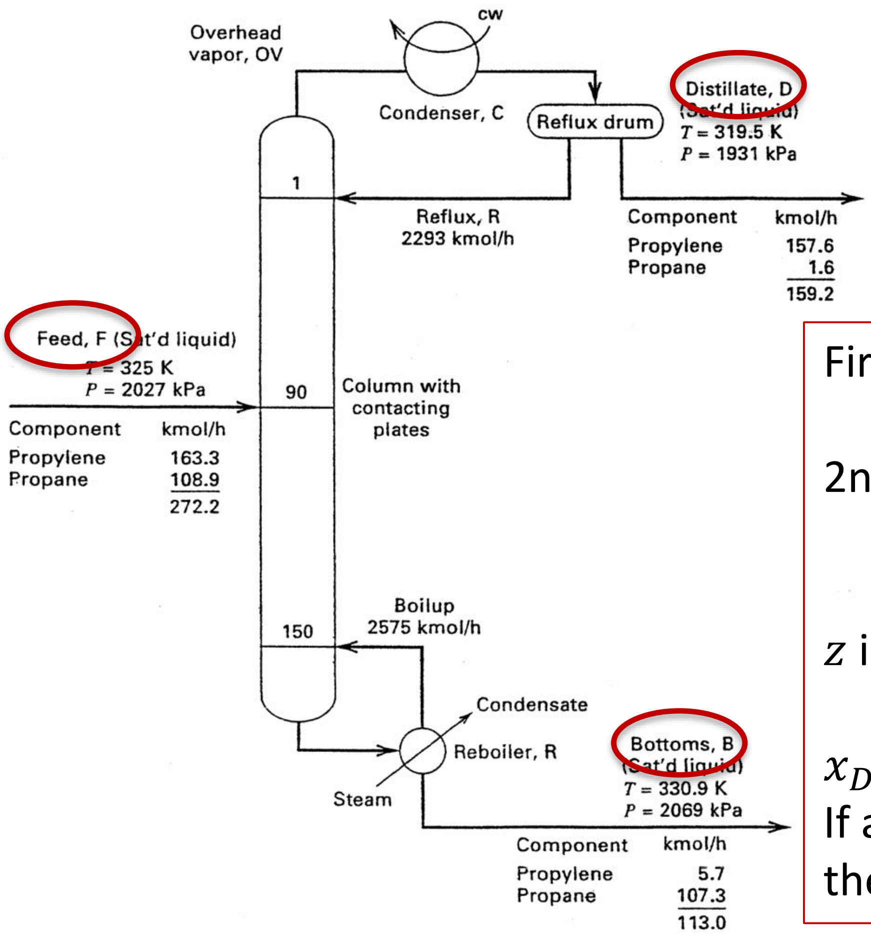
(1)Overall material balance
(2)Component balance.
(3)Then single equilibrium stage.
(4)Then the stages together.
(5)Possibly energy balances.

"Boil-up rate" is set by reboiler heating rate

View https://www.youtube.com/watch?v=I70jgRpf80o

First step is to calculate overall balances.

Initially, only some information is given, such as feed rate & composition and a product specification (purity or % recovery of desired species)



First balance: Total flows

$$F = D + B$$

2nd: Higher volatility species

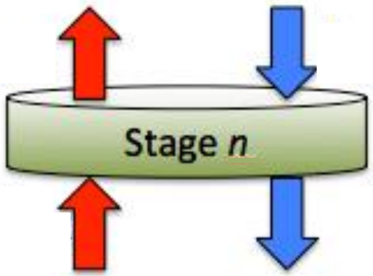
$$zF = x_D D + x_B B$$

z is feed mole fraction.

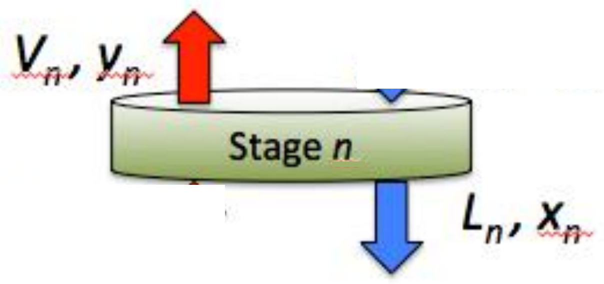
x_D may be specified product; If a "total condenser", same as the vapor from the top!

Fig. 1.16; Seader, Henley, and Roper, *Separation Process Principles*, 3rd Ed.

Then stagewise balances: Sketch a stage; label; write balances.

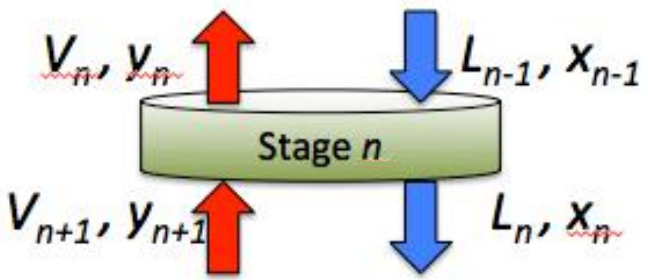


Sketch a stage; label; write balances.



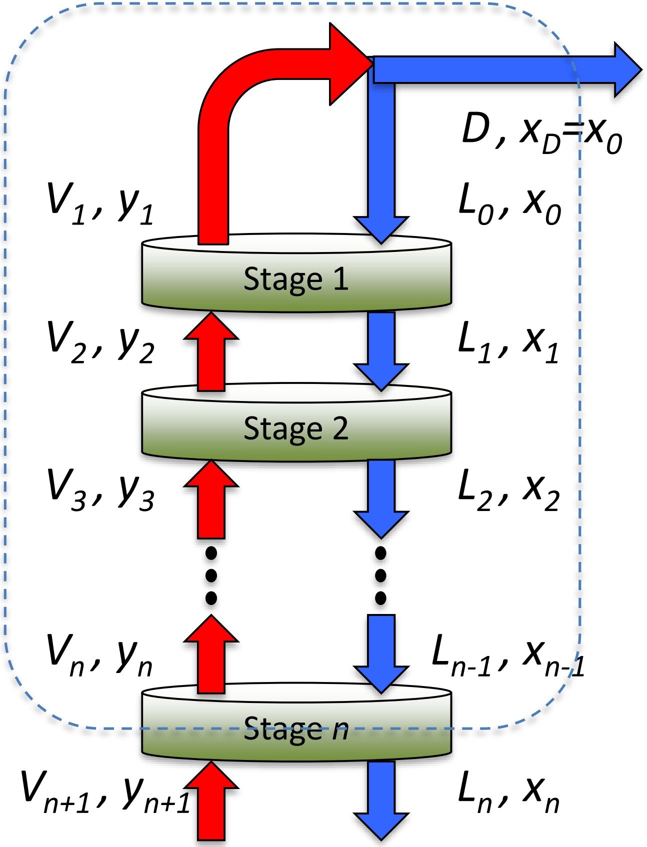
- Key point: Streams leaving stage *n* have subscripts *n*.
- Key point: Streams L_n and V_n are equilibrated (assumed).

Sketch a stage; label; write balances.



- Key point: Streams leaving stage *n* have subscripts *n*.
- Key point: Streams L_n and V_n are equilibrated (assumed).
- Number other streams from top or from bottom.
- We have to find the stage equilibrium compositions for all species, plus *T* and *P*.
- A classic approach is **McCabe-Thiele analysis** (like Theel'-ee).

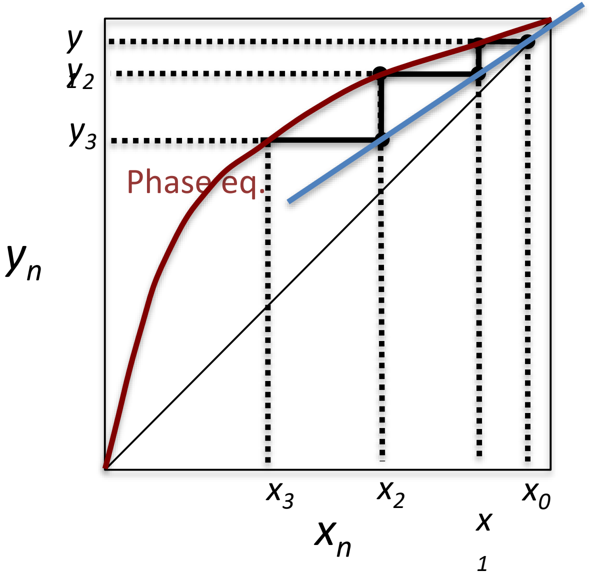
Analyze stages 1 thru n of enriching section: $R=L_0/D$, so $L_0/V_1 = R/(R+1)$.



$$V_{n+1}y_{n+1} = L_n x_n + D x_D$$

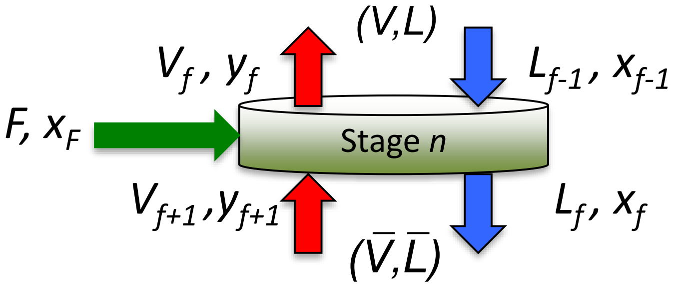
$$y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D x_D}{V_{n+1}} \quad \text{or} \quad y_{n+1} = \left(\frac{R}{R+1}\right) x_n + \left(\frac{x_D}{R+1}\right)$$

Equilibrium: $x_n = \dfrac{y_n}{\alpha + (1-\alpha) y_n}$



Feed stage of staged separation:

The rectifying-section O.L.
and the stripping O.L.
must intersect because both
must be satisfied at the feed stage.



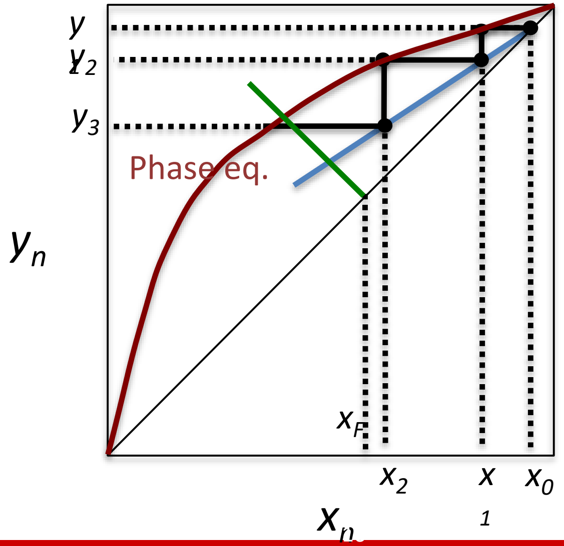
Solve together (with C.M.O.):

$$q = \frac{\bar{L} - L}{F} = 1 + \frac{\bar{V} - V}{F}$$

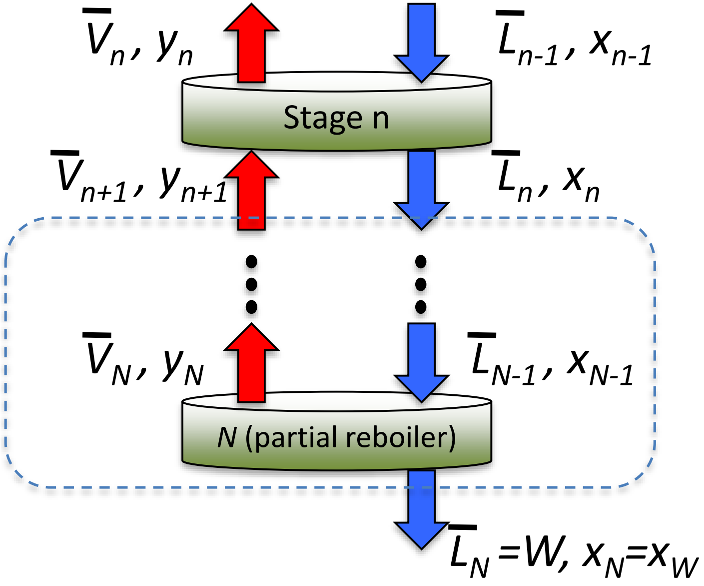
where q=(heat Q to reach sat'd vapor)/Q_{LV}

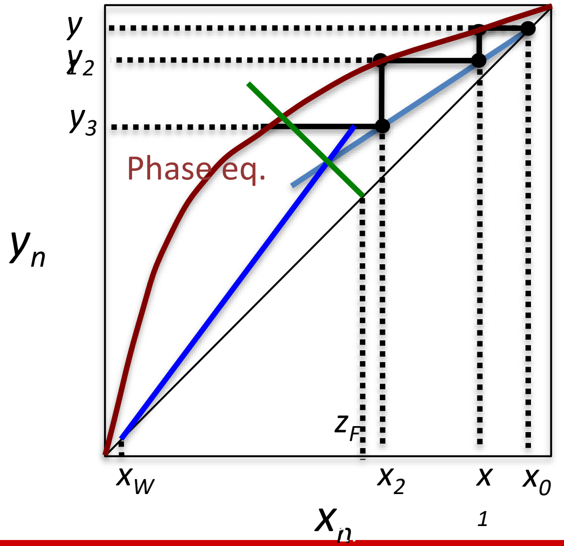
To get q line:

$$y = \frac{q}{q-1} x - \frac{1}{q-1} x_F$$



Stages *n*->*N* of stripping sectn; "V_B"=s=$V_N/\bar{W}$:





$$\bar{V}_{n+1} y_{n+1} + \bar{L}_N x_W = \bar{L}_n x_n$$

$$y_{n+1} = \frac{\bar{L}_n}{\bar{V}_{n+1}} x_n - \frac{W x_W}{\bar{V}_{n+1}} \quad \text{or} \quad y_{n+1} = \left(\frac{s+1}{s}\right) x_n - \left(\frac{x_W}{s}\right)$$

- Ideal equilibrium: $$y_A = \frac{\alpha_{AB} x_A}{1+(\alpha_{AB}-1)x_A} \qquad x_A = \frac{y_A}{y_A + \alpha_{AB}(1-y_A)}$$
  - Or empirical fit (non-ideal): $$y_A = \frac{a x_A}{1+(a-1)x_A} + b x_A (1-x_A)$$
- *q* line: $$y = \left(\frac{q}{q-1}\right)x - \left(\frac{x_F}{q-1}\right) \text{ where } q \equiv \frac{H_V - H_F}{H_V - H_L}$$
- Enriching-section operating line (**Constant molar overflow**): $$y_{n+1} = \left(\frac{L_n}{V_{n+1}}\right)x_n + \left(\frac{D x_D}{V_{n+1}}\right) = \left(\frac{R}{R+1}\right)x_n + \left(\frac{x_D}{R+1}\right)$$
- Stripping-section operating line (**Constant molar overflow**): $$y_{m+1} = \left(\frac{L_m}{V_{m+1}}\right)x_m - \left(\frac{W x_W}{V_{m+1}}\right) = \left(\frac{s+1}{s}\right)x_m - \left(\frac{x_W}{s}\right) \text{ where } s = \frac{\text{Boil-up rate or } V_{N+1}}{W}$$